Moodialogue: Transforming Emotions into Personified, Conversational Agents

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Abstract

Understanding emotion for self-awareness requires recognizing not only its type and intensity but also the surrounding context and the insights it provides. While prior work has studied emotion recording, little attention has given to how such records might foster reflection on context. To address this gap, we developed Moodialogue, a system that enables users to personify emotion and engage in dialogue with it. In a six-week field study with nine participants, we found that personified emotion records supported dialogues that uncovered overlooked contexts and coexisting feelings beyond a single entry. Participants also reported savoring positive emotions more deeply and reframing negative ones from new perspectives. These findings point to design opportunities for systems that move beyond recording, enabling post-recording interactions that deepen reflection on emotional context and meaning.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**;

Keywords

Journaling, Self-reflection, Emotion tracking, Field study, Design

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1 INTRODUCTION

Emotions play a central role in everyday life, influencing how individuals perceive events, make decisions, and understand themselves [62]. People often record their emotional experiences in various forms such as journaling [21, 41] or mood tracking [1, 26]. These practices can be flexible [4, 28] or structured depending on the purpose of mood recording, helping individuals reflect on what they feel and why. The act of recording emotions itself can support emotional well-being, as articulating feelings helps individuals clarify and externalize internal experiences [6, 28]. Through this process, users may gain self-understanding and engage in deeper self-reflection [8], and in some cases, the act of expressing emotions can contribute to emotional regulation [72]

More specifically, expressing emotions in one's own words or through abstract representations can help regulate them [26, 34]. However, simple mappings or numerical scales based on predefined emotion models may limit users' ability to fully articulate or resonate with their experiences [2, 22, 62]. This is because the interpretation of an emotion, even the same one, varies depending on how individuals perceive the emotion and the events that triggered it [9, 40, 62]. Thus, the perspective users adopt when describing emotions in their own terms is crucial for understanding those experiences.

Nevertheless, Schueller, et al. [62] pointed out that most mood-tracking apps currently available in the market fall short of helping users interpret their data or translate it into concrete strategies for improving mood. Since emotions involve not only behaviors but also unconscious cognitive and physiological reactions, it is difficult for users to regulate them on their own [58]. Prior work

has sought to complement users' subjective reflections through sensors [1, 55], photos [23], and reflective questions [24]. These tools connect objective data with subjective experiences and broaden opportunities for self-reflection. Yet, such information is not always directly tied to users' lived experiences [67]. Ultimately, users reconstruct these data into personally meaningful narratives [67]. This underscores the lasting importance of subjective interpretation in fostering emotional understanding and reflection.

At the same time, this personal interpretation process may at times lead users to confront unwanted realizations, which can have negative effects [20, 54, 67]. Kersten - van Dijk, et al. [25] emphasized that attending to one's inner thoughts and emotions in a reflective rather than ruminative manner is more adaptive. This aligns with prior studies that stressed the importance of moving beyond the emotions themselves toward considering their context or future plans [37, 76]. Accordingly, there is potential to design systems that help users observe the the causal structures surrounding their emotions from a certain distance. Such systems could reduce rumination and support the reinterpretation of events from new perspectives [63]. These insights suggest that reflection tools should help users express emotions in their own language. Moreover, they should also guide users beyond initial recordings toward deeper reflection and meaning reconstruction.

Before introducing Moodialogue, we clarify how the terms emotion and mood are used based on several literatures [15, 42, 61] . Emotion refers to an immediate, event-based response to a specific stimulus**.** Mood, in contrast, is a longer-term affective state that reflects one's overall emotional tone and balance. For example, when users personify their feelings, they engage with emotions tied to specific events; therefore, we refer to these as emotion agents. The name Moodialogue reflects its ability to accumulate such emotion-level records over time, helping users recognize and reflect on their broader mood states. In this study, we investigate this potential through Moodialogue (Figure 1), a mobile application that enables users to record their emotions, transform them into personified emotion agents, and engage in dialogue with these agents. More specifically, Moodialogue facilitates users' specification of the emotion type, the triggering context, a chosen name, and a visual profile image, which are then integrated to create a personified agent with whom they can engage in conversation to explore the situations in which that emotion arises. This study does not seek to directly assess the effectiveness of Moodialogue in improving emotional well-being. Rather, it aims to examine how users integrate Moodialogue into their everyday lives and what forms of emotional reflection emerge through its use. We focus on how the processes of visually personifying emotions and subsequently engaging in dialogue with them influenced participants' contextual understanding of emotions, self-reflection. For this we deployed Moodialogue on the personal mobile devices of nine participants and invited them to freely personify and converse with their emotions over the course of six weeks. During the study, we conducted 4 semi-structured interviews to how participants incorporated Moodialogue into their emotional reflection routines in everyday contexts. We evaluated the reflective experiences supported by *Moodialogue* and collected data, including interviews, chat logs, journals, and system logs. We then examined how personification and dialogue re-engagement helped participants contextualize and reconstruct emotional meaning.

The contributions of this work are threefold:

- **Design and implementation of Moodialogue**, which enables users to record and visually personify emotions, then reconstruct events through conversations with those personified emotions.
- **Empirical findings** from a six-week field study that reveal how dialogues with personified emotions influence users' reflection and understanding of emotional experiences.
- **Design implications for future emotion-support systems** include enabling users to reflect on the contextual meanings of their recorded emotions

Figure 1: Main screen of Moodialogue, which journals daily events through conversations with user-personified emotion agents.

2 RELATED WORK

The related work falls into strands (1) supporting user-defined emotional expression and (2) facilitating new perspectives for interpreting emotional experiences,

2.1 Supporting User-Defined Emotional Expression in HCI

One strategy used in HCI for accumulating users' emotional data is the use of self-reporting methods, which allow users to articulate their own emotions. Self-reports have been valued for helping users revisit their experiences and recognize recurring emotional patterns [22, 26, 34]. For this, appropriately determining recording topics and parameters is crucial. Lengthy measures may compromise usability [75], whereas overly minimal formats risk reducing the quality of data for subsequent analysis. To address this, Torkamaan and Ziegler [67] proposed an adaptive recording method that detects emotion changes and prompts additional questions to balance efficiency and data richness.

Several self-tracking applications [5, 35, 70] have explored such data for analytical purposes, aiming to integrate emotion and mental state monitoring into everyday life. Beyond analysis, prior work has emphasized the importance of enabling user-defined representations of emotion, since the same emotion can be interpreted differently across individuals [9, 40]. For example, Kim, et al. [28] proposed a method for expressing one's mood through single-stroke drawings, through a mobile application, and found that it helped users capture and express subtle affective nuances. In addition, Trackly [3] allowed users to customize visual trackers by mapping emotion parameters to pictorial elements. Users could also choose their own color schemes, maintaining agency in how emotions were represented. Outside of mobile contexts, other systems have supported user-defined expression through light [26] or physical form [34]. Collectively, these studies highlight both the importance of naming conventions that resonate with users [9] and the need for flexible, user-defined recording methods that can reflect subtle variations in emotion [2].

In addition, prior studies [11, 37] have emphasized that recording emotions should not only capture the classification or definition of the emotion, but also the triggers and circumstances under which they occur. Schueller, et al. [62], for instance, found through interviews with mood-tracking app users that they sought more than emotions records—they wanted interpretive cues such as causes and coping strategies. This highlights the need for an evocative balance [68], where emotion interactions help users recall experiences and reinterpret them to create lasting meaning. To address this issue, several studies have investigated ways to prompt users to explore their own emotional interpretation by providing cues such as physiological data. EmotionTuner [1], for example, combined PPG signal, GPS location, and user feedback, enabling users to identify stress triggers and record their associated feelings and thoughts. Similarly, EmoVis [55] visualized data collected from a wearable watch and offered a space for users to annotate their reflections. These approaches support journaling of subjective thoughts while simultaneously fostering emotional exploration and self-understanding [13]. However, since users must still interpret their own data, challenges remain: personal bias can shape emotional reflection [57], and some users may find interpretation difficult [62]. This underscores the need for systems that provide new perspectives and interpretive scaffolds—beyond recording—to help users make sense of their emotions in ways that generate sustained meaning.

In summary, prior work showed the potential of user-defined emotion recording to capture nuanced affective experiences. Yet, moving beyond simple recording toward tools that connect emotions with their triggers, coping mechanisms, and interpretive insights remains a critical design opportunity.

2.2 Facilitating New Perspectives for Objective Emotional Insight

Supporting objective thinking during emotional reflection is critical. Subjective interpretations can reinforce negative thought patterns, causing even neutral or positive events to be perceived negatively [57]. To counteract this, cognitive reappraisal—the reinterpretation of experiences from alternative perspectives—has been shown to be effective [73]. Thus, while it is important to respect users' subjective feelings when journaling, it is equally essential to complement them with tools that help recognize memory biases and distortions, thereby fostering more balanced reflection.

Recent advances in large language model (LLM)-based conversational interfaces have highlighted their potential to support such processes. LLMs can offer perspectives that users may not generate on their own. This facilitates cognitive restructuring, self-reflection, and critical thinking. For example, Tanprasert, et al. [69] addressed YouTube filter bubbles by presenting users with opposing perspectives, helping them broaden reflection and critical reasoning. Sharma, et al. [63] developed a system that detected cognitive distortions in user statements and offered alternative interpretations, supporting more adaptive thought processes. These studies suggest that LLMs can play an active role in enhancing emotional wellbeing by shaping how users evaluate and reflect on their experiences. In the context of self-journaling systems, LLMs have been used to lower the burden of writing and to deepen reflection by prompting users with new perspectives [29, 41]. For instance, *Diary Mate* [30] generated sentence suggestions based on user-provided keywords, simultaneously easing the effort of writing and encouraging users to reflect on experiences from diverse perspectives. Similarly, MindScape [41] supported reflective journaling by using digital logs to generate personalized prompts, helping users capture a wider range of emotional and narrative experiences. From a clinical perspective , MindfulDiary [29] adopted a conversational structure modeled on psychiatric interview protocols, enabling more systematic journaling of everyday experiences. Such approaches show that LLMs can function not just as passive recorders, but as mediators that help users rediscover overlooked emotions or perspectives and naturally refine the tone and nuance of their writing. However, Benharrak et al. [7] critiqued that agent personas often rely on minimal prompt descriptions, which might lead the model to reproduce persona traits in a generic, unspecific and repetitive manner. They instead proposed having writers define reader personas, shifting the focus from AI-centered assistance to the writer–reader relationship. This indicates that such a shift may also benefit emotional reflective journaling, particularly as little is known about how users interact with agent personas aligned with different emotional contexts.

In addition, research on the stance of conversational agents—a key component of agent persona—has been actively explored in the design of LLM-based reflective systems. Ryou, et al. [60] showed that simple empathic responses are not always effective; inappropriate or overly generic empathy can even undermine user experience. Similarly, Sharma, et al. [63] found that overly positive feedback or surface-level empathy may hinder reflection. In response, Kim, et al. [27] emphasized the value of open-ended questions that invite users to narrate their feelings and experiences more naturally, as opposed to closed, information-seeking prompts. When accompanied by contextually relevant probes, such open-ended scaffolding can simulate a dialogue with an attentive listener, fostering richer emotional expression and deeper reflection [49].

2.3 Summary and Reflection to Moodialogue

In summary, prior research has shown that enabling users to express emotions in their own language or to receive alternative perspectives can foster self-reflection. Such approaches extend emotion

recording beyond simple data accumulation, enabling users to adopt a more objective stance and reduce cognitive distortions. Recent work further highlights the potential of LLM-based interfaces to deepen reflection by supporting cognitive restructuring and offering new perspectives. However, existing studies have focused on recalling or reappraisal past emotions, without sufficiently addressing ways to reinterpret the contextual causes of emotions or to reframe them in relation to future. Moreover, within LLM-based emotional reflection systems, there is a lack of research exploring how AI agents might adopt diverse personas aligned with different emotional contexts, rather than functioning merely as assistants. To extend this gap, we propose Moodialogue, a system that allows users to engage in dialogue with personified emotion agents derived from their recorded emotions. By integrating user-driven expression with the discovery of contextual insights, Moodialogue offers a distinctive approach to enabling deeper emotional reflection and meaning-making.

In summary, prior research has shown that enabling users to express emotions in their own language or to encounter alternative perspectives can foster self-reflection. Such approaches extend emotion recording beyond simple data accumulation, allowing users to adopt a more reflective stance and reduce cognitive distortions. Recent work further highlights the potential of LLM-based interfaces to deepen reflection by supporting cognitive restructuring and offering new perspectives. However, existing studies have emphasized reflection during or immediately following recall or reappraisal, with comparatively less attention to how the meanings of emotions may evolve across contexts or be revisited over time. Moreover, within LLM-based emotional reflection systems, emotions are often engaged with as conversational content situated within a session, rather than as representations that can be repeatedly re-encountered and reflected on across different situations. To address this gap, we propose Moodialogue, a system that treats recorded emotions as externalized, interactional entities that users can revisit and reflect on through dialogue over time. By integrating user-driven expression with ongoing contextual reinterpretation, Moodialogue offers a distinctive approach to supporting deeper emotional reflection and meaning-making.

3 BACKGROUND DESIGN RATIONALE, AND IMPLEMENTATION OF MOODIALOGUE

Moodialogue is a research product [46] designed to support journaling through the process of personification of felt emotion and engaging in dialogue with them. The system was developed in response to the inquiry into how emotion personification and emotion-based dialogue in everyday contexts influence users' emotional awareness and reflection. To this end, Moodialogue was built with sufficient quality, finish, and independence to be deployed in participants' real-life environments over an extended period without researcher intervention. By implementing and distributing Moodialogue in situ, we sought to ensure that participants would perceive it as a real product and to observe how their experiences with it evolved over time [47].

3.1 Background and Design Rationale

Moodialogue consists of two core features. First, the personification process, in which users classify, label, and visually represent their emotions. Second, the dialoguing and journaling process enables users to converse with their labeled emotions about related events. Through these conversations, they gain reflective insights and transform their interactions into journal entries.

3.1.1 Emotion Record and Personification Process. The first design feature of Moodialogue is user-driven emotion record and personification process. Prior work shows that articulating emotions helps individuals understand their feelings and can itself have a regulatory effect, even without intentional control [14, 22, 36]. However, fixed models such as Plutchik's wheel risk limit personal resonance, since interpretations of the same emotion vary across individuals [2, 9, 22, 40, 62]. However, unstructured emotion articulation can be cognitively demanding, particularly when emotions are mixed or ambiguous and lack clear verbal labels [65]. To balance these tensions, *Moodialogue* presents Plutchik's eight basic emotions [51] as an initial orienting scaffold. Rather than prescribing the emotion, these categories function as a cue that helps users find a starting point.

The emotion record and personification process consist of the following steps. First, the user selects one of the eight basic emotions from Plutchik's wheel and describes its details in their own words (Figure 2a). During the design of Moodialogue, we considered multiple emotion models as potential ways of structuring emotion records, including Russell's circumplex model and Ekman's categorical approach. Russell's circumplex model represents emotions through valence–arousal coordinates [59], and Ekman's model categorizes them based on facial expressions [12]. In contrast, Plutchik's model [50] offers eight emotion groups—joy, trust, fear, surprise, sadness, disgust, anger, and anticipation—that are more intuitively recognizable in everyday life. We adopted this model because it offers a concise and accessible set of basic emotions. Selecting a basic emotion first serves as a cognitive scaffold, allowing users to establish an initial emotional direction and then reconstruct finer nuances through their own labels and contextual descriptions. Next, the user sets a preference for that emotion (Figure 2b) and assigns it a name (Figure 2c). After defining this basic information, the user creates a visual profile of the emotion by selecting from several shapes—polygon, line, circle, or star (Figure 2d-1). Rather than using precise tools that force users to express emotions in free forms, we aimed to provide interpretive openness by allowing more visually ambiguous representations, similar to Kim, et al. [28]'s research. Based on this intention, we chose to let users compose their expressions using minimal and familiar units—lines, polygons, circles, and stars—by generating corresponding profile images for the recorded emotions (Figure 2d-2). After creating a profile image, the user can record contextual details of the emotion, such as triggering situations (Figure 2e), situations in which the emotion fades (Figure 2f), and their reflections on the reasons for its occurrence (Figure 2g). Once the user completes the process, the new emotion is recorded, and Moodialogue generates a persona for the corresponding emotion agent based on the recorded information for future conversations.

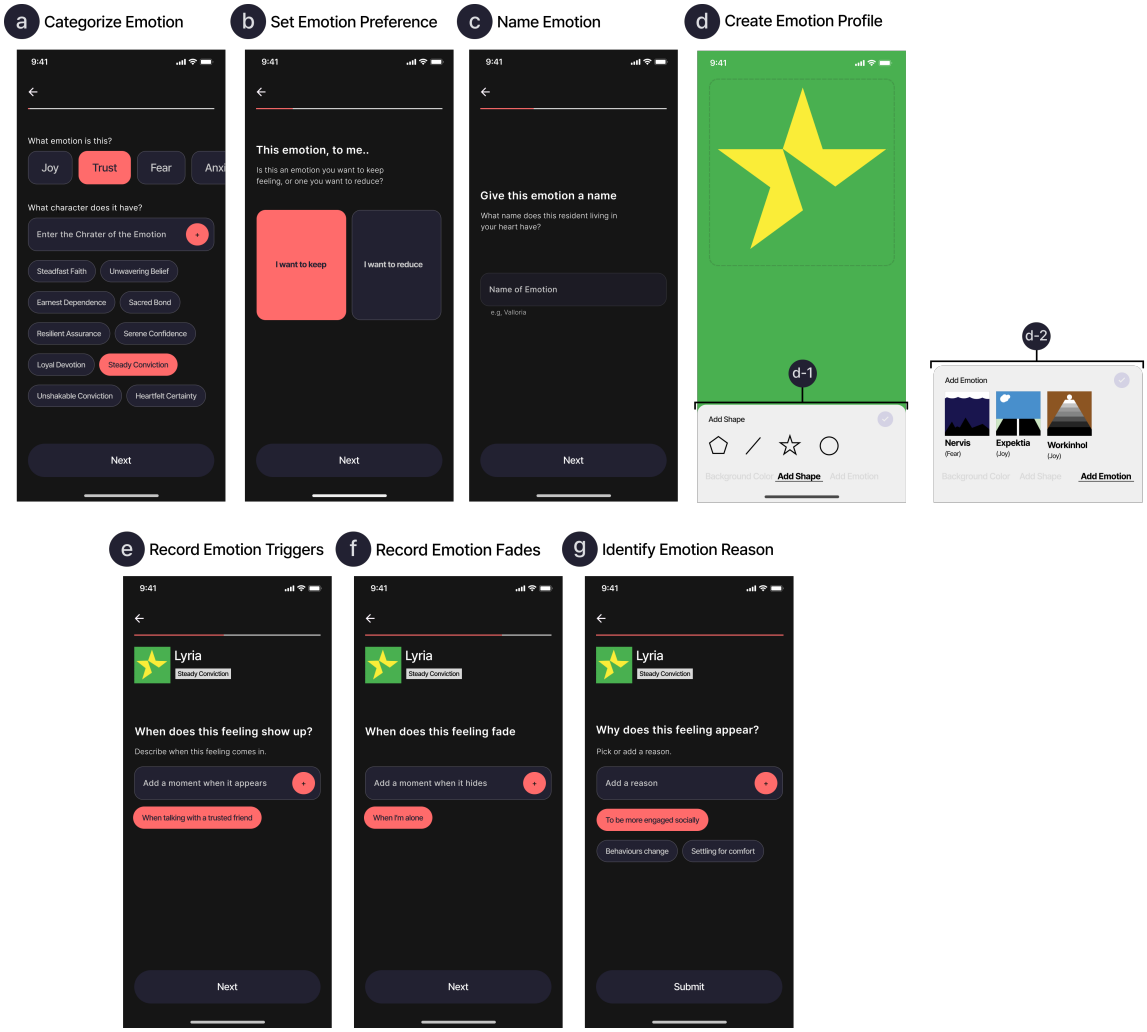

Figure 2: Emotion personification in Moodialogue

3.1.2 Dialoguing with Personified Emotions and Converting into Journals. When collecting personal data, it should be viewed not just as numerical indicators but as contextual information that supports broader self-reflection rather than isolated analysis [39]. In the context of emotions, Kim, et al. [28] emphasized the need to move beyond simple collection toward approaches that expand recorded emotions to provide deeper insights. Building on this, we designed Moodialogue to extend emotional experiences by assigning a "persona" to recorded emotions and enabling users to engage in dialogue with them.

Through this design, and in line with prior HCI research [29, 30], users can reinterpret daily events from multiple perspectives and uncover aspects of experiences or feelings that might otherwise be overlooked. The system also aims to facilitate cognitive reframing of negative thoughts [63] and promote more positive perspectives [41]. To achieve this, the interlocutor is framed not as a generic conversational agent but as the emotion itself, applying the psychological strategy of *distanced self-talk* [17, 48]. Distanced self-talk refers to the practice of addressing oneself from a third-person perspective (e.g., using one's own name rather than "I"), which has been shown to help individuals create psychological distance from emotional experiences [17]. This framing encourages users to reflect on events more rationally [32] and objectively [31, 48], while creating a sense of psychological distance [17] that supports self-reflection. Building on this concept, we aimed to encourage users to address their emotions from a third-person perspective, allowing them to observe their feelings with psychological distance. Moreover, extending prior LLM-based journaling systems that position the conversational agent as supporting self-reflection (e.g., Diary Mate {Kim, 2024 #1128}, MindfulDiary {Kim, 2024 #1037}, MindScape {Nepal, 2024 #1184}), Moodialogue treats recorded emotions as externalized, interactional entities that users can engage with over time. By structurally embedding distanced self-talk [17, 48] at the interaction level, this design makes psychological distance a default condition of reflection rather than an optional conversational strategy.

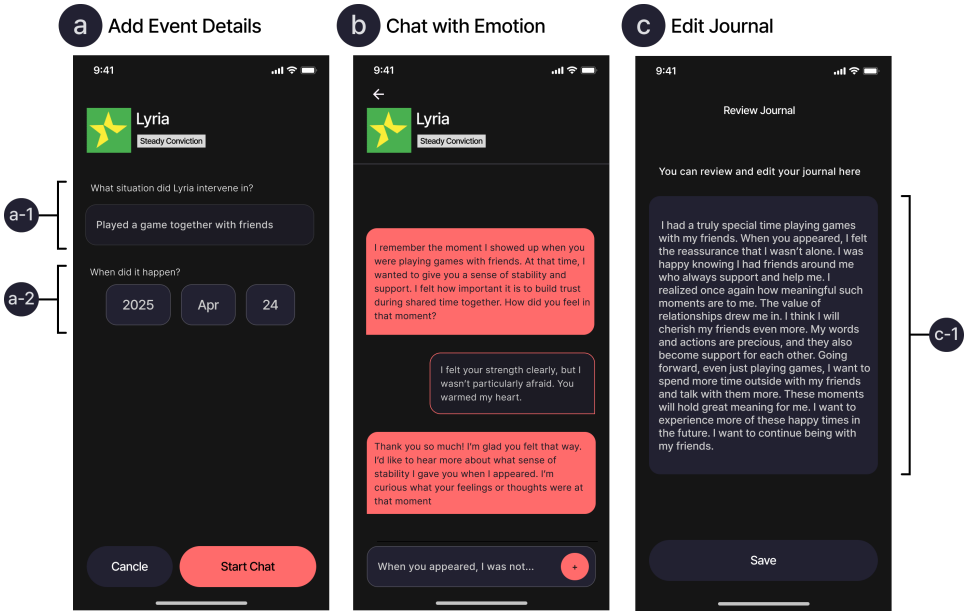

Figure 3: Journaling process of Moodialogue

The journaling process begins with the user selecting a personified emotion experienced during the day. Once the user selects a personified emotion, they can record the name of an event they wish to discuss (Figure 3a-1) along with the date of that event (Figure 3a-2). Based on the persona of the emotion and the user-provided event name, Moodialogue creates a chat room where the user can engage in dialogue with the personified emotion. The emotion agent initiates the conversation and poses follow-up questions according to the user's responses (Figure 3b). After four or more conversational turns, the user may end the session, at which point Moodialogue generates a journal draft based on the conversation history. This draft can then be refined and saved by the user as a final entry (Figure 3c). Importantly, conversations with a personified emotion are not limited to a single session; users may revisit the same emotion later to initiate new dialogues and deepen their reflection.

3.2 Implementation of Moodialogue

Moodialogue consists of a cross-platform mobile application, an emotion-based chatbot API, and a Supabase-managed PostgreSQL database. This app is act as a research product [46], which is built for user study purpose and not publish to app market

Application and Data Storage. The mobile app, developed with Flutter for both iOS and Android, allows users to create visual emotion profiles through customizable shapes and colors using the *CustomPainter* widget. Distribution was handled via TestFlight (iOS) and APK installation (Android). User authentication, emotion profiles, and chat logs are securely managed through Supabase authentication and access control, with all data stored in a PostgreSQL database.

Chat Agent and API. The conversational agent was implemented as a LangChain-based pipeline and deployed with FastAPI (LLM: gpt-4o-mini, temperature: 0.3). When users record a new emotion, the information is passed to the persona-generation prompt, which automatically constructs the corresponding emotion persona. This process does not rely solely on user-defined attributes. It prompts the LLM to infer both explicit and latent emotional characteristics, creating a nuanced profile that may include aspects users have not consciously recognized. The generated persona then serves as the basis for the emotion agent's personality and manner of expression during subsequent dialogues. Regarding process of chat session, Figure 4 illustrates the integrated process of chat sessions and diary generation. During the initial stage (within the first three turns), the dialogue focuses on contextual questions about the event, with the *answerGenerator* producing exploratory prompts while the *responseStrategyPlanner* remains inactive. Once the conversation extends to four or more turns, the *responseStrategyPlanner* intervenes to guide the subsequent flow, and the *AnswerGenerator* generates follow-up questions that address the causes and interpretations of the emotion. At this stage, the system transitions to the diary chain, where the *diaryUpdater* creates a draft entry based on the accumulated dialogue. This draft is then reviewed by the *diaryEvaluator*, which filters out unsupported or redundant content to ensure consistency. The refined output is finalized as a diary draft, which can be continuously updated as new user responses and subsequent dialogues are added. This chained structure enables the system to adapt to user needs while transforming conversations into coherent and evolving records of reflection.

4 FIELD STUDY

We conducted a six-week field study with nine participants to investigate how Moodialogue was used in everyday life for emotion recording, and dialogue-based emotional reflection. Specifically, we examined how the processes of visually personifying emotions

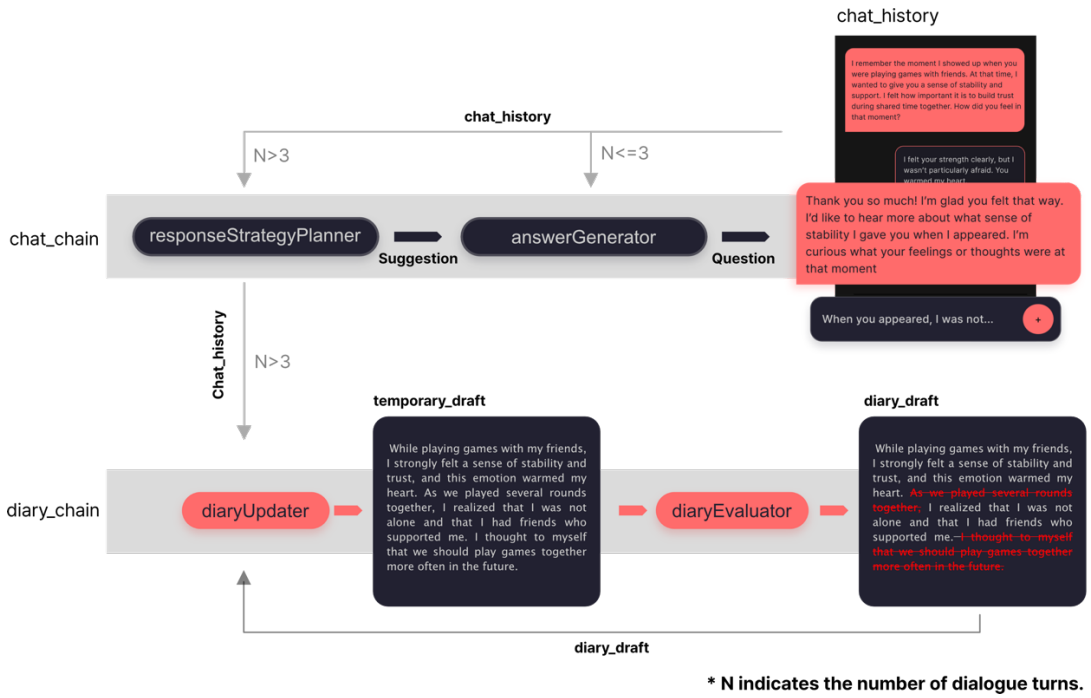

Figure 4: Moodialogue's process chains for chat and diary draft creation. The upper chain represents the process of generating responses for the personified emotion, while the lower chain represents the process of creating a diary draft based on the chat history.

and subsequently engaging in dialogue with them influenced participants' contextual understanding of emotions, self-reflection.

4.1 Participants and Installation

Participants were recruited through local community boards and campus social media channels at two universities, targeting individuals who frequently record their emotional experience (e.g., journal, emotion tracking). To screen for suitability, we distributed a Google Form survey that included questions about prior journaling experience, emotion-tracking practices, perceived emotional awareness, challenges in maintaining journaling, and prior use of AI tools.

We focused not only on participants' prior self-tracking habits but also on their emotional awareness—how confidently they recognized and articulated their feelings—and the challenges they faced during reflection. These factors shape how individuals engage with Moodialogue's emotional personification, influencing whether the agent's prompts help clarify what to express or instead intensify discomfort. Understanding these baseline tendencies was therefore crucial for interpreting each participant's reflective process. We also sought to understand the participants' prior familiarity with LLM-based tools by examining their previous experiences with AI technologies. From 29 applicants, we selected participants based on the following criteria: (1) no reluctance toward recording emotions or emotional experiences, (2) the ability and willingness to reflect on emotions in daily life, and (3) consistent engagement with recording emotional experience practices. A total of nine participants were selected for the study. Although the small sample size may limit the generalizability of our findings, prior research [44, 45, 52] has highlighted the value of conducting qualitative, in-the-wild investigations with small participant groups. These studies emphasize observing experiences within everyday contexts over extended periods. In line with this approach, we focused on a small selection of participants to apprehend the nuanced, contextual, and layered nature of emotional experiences in depth. The detailed demographic information of each participant is described in Table1:

Table 1: Details of each participant

Code	Gender/Age	Recording Method	Frequency
P1	M/21	Social media	1–2 / week
P2	F/21	Paper diary	1–2 / week
P3	F/24	Diary app	Daily
P4	M/32	Paper diary	1–2 / week
P5	F/30	Paper diary, apps	3–5 / week
P6	F/26	Paper diary	1–2 / week
P7	M/19	Diary app	3–5 / week
P8	M/20	Mood tracking app	Daily
P9	F/20	Diary & Mood app	1–2 / week

Five of the nine participants (P1, P2, P5, P7, P9) reported a high level of self-reported emotion awareness competency, noting that they could recognize and articulate their emotions relatively clearly. They attributed this to their tendency to take an objective stance

toward emotions (P1), strong verbal clarity when describing feelings (P2), feedback from others that they express emotions well (P5), or established habits of daily reflection (P7, P9).

Three participants (P3, P6, P8) reported a moderate level. They stated that while they could generally identify their emotions, they required additional effort to organize or verbalize them (P6). Some noted that their emotions became ambiguous depending on the situation or faded over time (P8), or that their expressions lacked consistency (P3). One participant (P4) reported a low level, explaining that their emotions did not take on clear forms or that defining them felt inherently difficult.

Participants also described several challenges in emotional reflection. Some (P1, P2, P9) found it difficult to separate and organize multiple emotions occurring at the same time. P4 and P5 noted discomfort or overthinking when recalling and narrating past emotions. P6 and P7 reported that everyday time constraints made it difficult to maintain reflective habits. P8 explained that emotions often faded with time, making accurate recording challenging.

Installation. Prior to the study, participants were provided with a user manual and installation guide for *Moodialogue*. Android users installed the application via an APK file, while iOS users accessed it through TestFlight. The field study ran for six weeks in total. One day before the study began, we confirmed that each participant had successfully installed the application and conducted a brief demo session to familiarize them with its main design features and ensure usage during the study. The demo session lasted approximately 15 minutes per participant. The researcher installed Moodialogue on their smartphone and shared the screen on a monitor so that participants could follow each step. During the demo, the researcher introduced the main features of Moodialogue—including emotion personification, conversations with emotion agents, and journal transformation. Intermittently we asked questions to ensure that participants fully understood the functions and overall usage flow.

Ethical Considerations: This study was approved by the Institutional Review Board (KAISTIRB-2025-255). A crisis-response protocol was prepared to immediately terminate the study and refer participants to professional counseling services if they reported discomfort or emotional distress during the stdudy. Before participation, we also informed participants of the potential risks associated with LLM-based agents—such as model bias and the possibility of hallucinated outputs—and explained how their data would be processed. Moodialogue uses the OpenAI API, which does not use user data for model training[1] and retains it on servers for a maximum of 30 days[2] under its privacy policy. To further minimize re-identification risks, Moodialogue transmits only de-identified conversation text—without any personally identifiable information—when making API calls.

To mitigate potential biases of the AI model, we designed a dual-prompt architecture that separates the roles of the *ResponseStrategyPlanner* and the *AnswerGenerator*. The *ResponseStrategyPlanner* models the stance of a clinical psychologist, generating empathy-focused conversational strategies. The *AnswerGenerator* is restricted to responding solely in a first-person perspective that represents the user's emotion, preventing the model from adopting an advisory or authoritative role. Even among the strategies proposed by the *ResponseStrategyPlanner*, only empathic and reflective elements are selectively applied to avoid diagnostic or directive language. This architecture acts as a safety mechanism that prioritizes emotional validation and reduces the risk of the model being mistaken for a counselor.

[1]https://help.openai.com/en/articles/5722486-how-your-data-is-used-to-improve-model-performance
[2]https://openai.com/ko-KR/policies/row-privacy-policy/

4.2 Data Collection, and Analysis

Interview Data. Over the course of the six-week study, we conducted four interviews with each participant (at Weeks 2, 3, 4, and 6), each lasting approximately 40 minutes. Each session focused on a different theme, as summarized in Table 2, and subsequent interviews incorporated follow-up questions tailored to participants' earlier responses. All interviews were recorded and transcribed. Based on the transcripts, we generated open codes. To ensure triangulation, two more researchers who were not directly involved in the study independently conducted the coding. Each researcher independently reviewed all interview transcripts and generated open codes, aiming to capture not only recurring patterns but also participants' distinctive, situated experiences. During this stage, researchers did not share or discuss their codes to avoid premature convergence and to preserve the diversity of interpretive perspectives. After individual coding was completed, the research team engaged in collaborative discussions to consolidate and refine the codes. Through this iterative process, an initial set of 11 codes was synthesized into 7 overarching themes by identifying conceptual relationships and clustering related insights. Figure 5 illustrates the overall interview analysis process.

Chat and Journal Data. We analyzed the database logs of participants' interactions with Moodialogue. This included both chat history with personified emotions and the journal entries generated from those dialogues. The analysis focused on the following log data: the time of emotion records, conversation start times, number of turns, number of new emotions per week, and frequency of revisiting past personified emotions. For the chat and journal data, we analyzed only the diary entries that participants consented to share, excluding any personal or sensitive records to protect privacy. These materials were used as supplementary evidence to enrich and support the findings derived from the interviews. After reviewing all chat logs, we mapped relevant excerpts to the corresponding interview themes. This process enabled us to contextualize participants' reflections in relation to the dialogues they had with their personified emotion agents. This process provided deeper interpretive insight that could not be fully understood from interview data alone.

5 FINDINGS

During the field study, a total of 226 dialogue sessions were created. On average, each dialogue session consisted of 10.32 conversational turns. In total, 125 emotions were personified, comprising 70 positive and 55 negative emotions. The number of newly registered emotion agents per week was 33, 25, 20, 16, 13, and 18, respectively. In addition, participants also engaged in dialogue with previously registered emotions, rather than only with those personified on the same day. The weekly frequencies of such re-engagements were 9, 22, 15, 22, 21, and 29, whereas dialogues with emotions

Table 2: Interview topics by weeks

Week	Main Interview contents
First Interview (Week2)	Differences compared to previous experiences with emotion recording Which emotions were personified Initial experiences of dialogue with personified emotion agents
Second Interview (Week3)	Desired attitudes or questions during the dialogue process Effects of dialogue with personified emotion agents Insights gained from the dialogue process
Third Interview (Week4)	Differences in experiences across types of emotions Suitable user types and the role of personification and dialogue Changes in evaluation of Moodialogue compared to the beginning of the study
Fourth Interview (Week6)	Overall experiences throughout the study Feedback on improvements to Moodialogue's design features Reasons for the chosen emotion names and visual profile images

Description	Codes
When users primarily engaged with Moodialogue	**C1)** Night or Next-Morning Records
What role visual emotion profiles played in helping user recall and reflect on their emotions	**C2)** Quick Classification via Visual Profies of Emotions
How users decided which emotions to personify	**C3)** Recording Strongest Daily Emotion
How personifying and conversing with positive emotions influenced users	**C4)** Savoring Positive Emotions
How personifying and conversing with negative emotions influenced users	**C5)** Relief from Negative Emotions
How conversing with personified emotions helped users understand the situations in which those emotions arose	**C6)** Identifying Emotion Causes
What new insights users discovered through dialogues with personified emotions	**C7)** Discovering Coexisting Emotions
Users' needs for recording not only strong emotions but also the integrated flow and changes of emotions across a day	**C8)** Need to Track Emotional Changes
Users' needs for updating emotions personas with new insights	**C9)** Need to Update Recorded Personas
Users' needs for representing mixed or ambivalent emotions	**C10)** Need to Express Mixed Emotions
How users leveraged their own questions to supplement or reinterpret agent prompts	**C11)** User-Driven Questions in Dialogue

Themes	Findings
Defining and Recognizing Emotions through Personification **(C2, C3, C5, C8)**	**5.1 Enhancing Awareness and Expression of Emotions through Personification**
Gaining Immediacy in Emotion Recognition by Creating Visual Profile Image of Emotions **(C2,C10)**	
Recognizing Hidden Emotional Causes and Reinterpreting Experiences **(C4, C5, C6)**	**5.2 Becoming aware of a blind spot of Emotion through Chat with Personified Emotion**
Uncovering Coexisting Emotions beyond the Chosen Label **(C6, C7, C8, C10, C11)**	
Overall Usage and Evaluation of Moodialogue **(C1, C3)**	**General Finding**
Effects of Dialogue with Emotion Agents **(C4, C5, C9)**	
Roles of Mood Recording and Conversational Engagement **(C4, C5)**	

Figure 5: Comprehensive data analysis process

registered on the same day occurred 29, 18, 18, 16, 9, and 18 times. Regarding this, in the third-week interview, P8 noted, "This week, I found myself engaging more often in conversations with previously registered emotions, and the accumulation of these emotions made the experience feel more comfortable." In the same vein, P7 reflected in the final interview, "At first, registering a new emotion felt time-consuming. But once I could revisit and reuse the emotions I had already recorded, the process shifted from a one-time effort to something reusable, letting me express my emotions in a deeper and more concrete way."

Participants typically used Moodialogue during daily reflection, often before bedtime. They rarely used it immediately after an emotional event, but sometimes during free moments when emotions resurfaced. For example, P9 explained: *"Usually, before going to bed, I would recall the most memorable event of the day, talk with the strongest emotion I felt."* Similarly, P5 noted: *"I didn't have a fixed time. At first, I mostly did it before sleeping, but sometimes I started recording in the morning or afternoon as well."* At the same time, participants found limits in the app's single event-focused design. By focusing on a single emotion at a given moment,, it was difficult to capture the full range of emotions experienced throughout the day. P8 pointed out: *"If it's a diary, I could be able to record multiple emotions across the flow of a day—morning, lunch, evening—but now it's limited to a single event."* P7 added: *"Even if I try to include another emotion in the middle of a dialogue, it gets buried in the overall context. It would help if I could add short comments, like annotations."*

Moreover, Moodialogue encouraged users to move beyond one-time emotion recording. By allowing them to personify emotions and engage in repeated dialogues, it supported sustained reflection on specific emotional experiences over time. Participants described how the process of creating an emotion persona and conversing with it helped them confront their emotions more honestly and explore them in greater depth. As P8 noted: *"It didn't just end with 'I felt this emotion.' I thought more specifically about why I felt that way by recalling past events."* More specifically, participants emphasized that personification and dialogue played complementary roles.

The emotion personification process, combined with visualizing emotions as profile images, heightened users' awareness of their behaviors and emotional states. Thereby, the subsequent dialogues offered comfort, insights, and a means to interpret surrounding events. P3 explained: *"Both processes are necessary. Recording makes me revisit emotions I might have overlooked in my actions, and dialogue gives me comfort and helps me organize them."* Similarly, P1 reflected: *"Recording emotions is a process of organizing, while dialogue—especially with negative emotions—helped me gain lessons from them."* P7 added: *"Creating an emotion requires concrete thinking, but it's also expressive, like quickly generating a character from Inside Out. Dialogue, on the other hand, feels like adding more context, such as how that emotion came to me."* However, some participants felt that the system fell short in conveying a sense of ongoing interaction with the same emotion over time. P6 suggested: *"When I talk with an emotion, it would be helpful to bring up the past event when I felt it. Even if it's the same 'dopamine,' (name of P6's emotion) knowing how it differed from the day before would make it feel more like I'm talking with the same emotion."* Others expressed a desire to update emotion personas with new insights gained through dialogue. P8 remarked: *"If I could edit the emotion profile, I would add what I realized through the conversation. And over time, if it could automatically branch into more fine-grained emotions, it would become more concrete."*

5.1 Enhancing Awareness and Expression of Emotions through Personification

5.1.1 Defining and Recognizing Emotions through Personification. Participants perceived Moodialogue's process of personifying emotions not merely as a means of recording emotions, but as an act of classifying and defining them in their own way. P2 explained: *"Before chatting with the emotion, I preferred to first classify, describe, and visually express it. That way I understood the emotion more clearly before moving on to dialogue. If I had started with chatting, the emotion would have felt too abstract."* Similarly, P4 reflected: "*In the past, I would simply think, 'I felt this emotion.' But when using Moodialogue, I thought instead, 'Oh, this emotion came up in this kind of situation.'*" For some participants, recording negative emotions also contributed to relief and emotional release. P1 noted: *"Since recording takes some time, I end up reflecting more deeply. Especially for negative emotions—while not all of them are resolved—some, like fear, seem to ease as I carefully record them."* Likewise, P2 and P3 reported that visually expressing emotions helped them step back and diffuse anger: *"Decorating and visualizing the emotion made me see it from a distance, which helped me calm down."* In contrast, P8 described the prolonged expression process of negative emotions as frustrating: *"Spending too much time labeling negative emotions just made me annoyed and uncomfortable. I wanted to finish quickly."*

Participants also noted that Moodialogue encouraged them to focus on the day's most dominant emotion, often tied to a specific event. While this focus enhanced clarity and depth, it simultaneously constrained their ability to capture the fluidity and interplay of multiple emotions experienced throughout the day. Regarding this, P8 commented: "In a diary, I should be able to record several emotions across the day—morning, afternoon, evening—but now it feels restricted to just one event." Similarly, P6 remarked: "When I reflect on the whole day, I tend to miss small moments of irritation or stress. If I could record them briefly by time of day, I wouldn't lose them." These observations suggest that Moodialogue may underrepresent fleeting or time-specific emotions. Another theme concerned the long-term impact of revisiting personified emotions. While positive emotions acted as uplifting reminders, negative emotions sometimes triggered unwanted recollections. P5 shared: "Seeing my emotion profile made me recall the embarrassing and sad event. It felt like going back to that moment." P4 similarly described: "Looking back, I thought, 'I guess I don't record many positive things,' and it reminded me of recent negative experiences." Although most participants did not experience direct harm, these reflections point to potential emotional burdens of re-encountering negative emotions. To mitigate this, participants suggested ways to evolve visual profile images over time. P3 remarked: "Looking at the negative emotion profile gave me a strange feeling—it automatically brought back that past situation. If the profile changed color over time, it might reduce that kind of recall."

5.1.2 Gaining Immediacy in Emotion Recognition by Creating Visual Profile Image of Emotions. Seven participants reported that expressing emotions through colors and shapes helped them distinguish emotions more clearly and recall them more quickly when revisiting past records. P7 explained: *"Using colored lines is like a process of shaping emotions that text alone cannot fully capture—it gives me an immediate impression at a glance. Of course, it lacks detailed context. On the other hand, text can describe emotions in detail, but you don't really grasp the emotion until you read it."* Similarly, P8 noted: *"To explain a happy emotion in words, I would need a long conversation. But with visualization, I could capture it instantly. Using bright colors to represent pleasant memories, I immediately understood that I was happy."* In addition, visual profile images not only helped distinguish emotions at the time of recording but also allowed participants to connect and layer emotional nuances across time. For example, P5 described representing fear as *"a tingling anxiety"* using the image of a sea urchin, embedding within it a blend of other profiles representing sadness, fear, and anticipation—thus surfacing the layered nature of complex emotions (Figure 6a). In a similar context, P7 was found to use previous emotion profiles to express subtle nuances within the same emotion (Figure 6b). For instance, he extended the profile of sadness to represent *"being drawn into sadness,"* or to articulate two moments of regret: questioning *"Why did I start this?"* before beginning a task, and *"Why was I so tense?"* after completing it. In a slightly different context, P9 reflected that visual personification enabled her to bring past emotions into the present without discomfort (Figure 6c): *"If I write in text, it's hard to mention both—like being happy today but depressed*

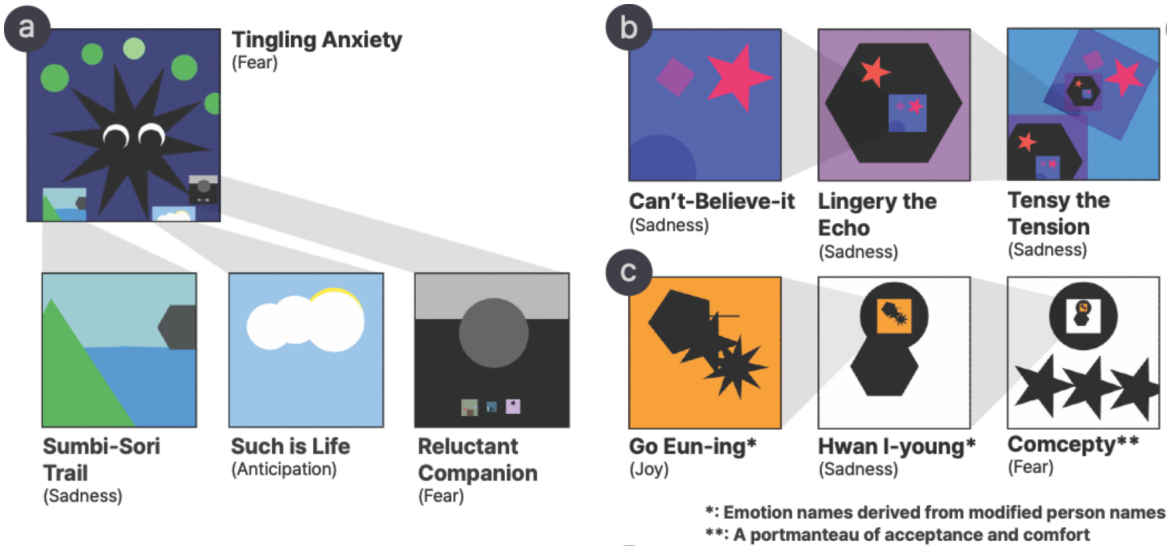

Figure 6: Examples of emotion profiles created by reusing previous profiles: (a) P5's "tingling anxiety," expressed through nuances of an earlier profile; (b) P7's profile, showing subtle variations of sadness, including being drawn into sadness and experiencing multiple sadness (both before and after an event); (c) P9's profile, reusing a previous day's emotion image to reveal unnoticed influences on the current emotion.

yesterday. But with an image, since it doesn't state emotions explicitly, I can carry yesterday's emotions into today's representation, showing that today's happiness may have grown out of yesterday's sadness." At the same time, not all participants found it easy to represent multiple or mixed emotions through Moodialogue. Both P1 and P5 expressed difficulty in capturing ambivalent or blended emotions. P5 noted: *"Sometimes joy is mixed with anger. Recording helps me realize that, but when I try to choose an emotion before I'm used to it, I feel constrained."* Similarly, P1 explained: *"When I felt mixed emotions, I prioritized the strongest one and recorded that. It was hard to capture them all at once. If I could record complex emotions together, it would allow for richer descriptions."* These mentions indicate that Moodialogue's personification process enhanced users' recognition and recall of emotions. However, by emphasizing dominant emotions, it may have constrained the expression of subtler or mixed emotional states.

5.2 Becoming Aware of a Blind Spot of Emotion through Chat with Personified Emotion

Engaging in dialogue with personified emotions helped participants view their emotions from a broader perspective and uncover blind spots that had previously been overlooked. Positive emotions were often revisited within their original contexts, enabling participants to savor those experiences. In contrast, negative emotions led participants to re-evaluate the circumstances of their occurrence and move beyond narrow or initial interpretations. Several participants (P1, P2, P3, P4, P8) commonly described this as an opportunity to detach from initial reactions and reinterpret their experiences. For example, P6 reflected that even small experiences became meaningful when revisited: *"Through recording, I could think back—'this is why I was excited, this is why I felt calm'—and that was enjoyable."* He further added that negative emotions, when reviewed and organized, could be reframed as learning opportunities: *"If I could look back and say, 'I learned this through that emotion,' it actually contributed to my growth."*

5.2.1 Recognizing Hidden Emotional Causes and Reinterpreting Experiences. Dialogues with personified emotion agents encouraged participants to re-examine the actions and contexts that triggered their emotions. Through this process, participants engaged in deeper reflection, often uncovering situational nuances that had previously been overlooked. P2 described: "When the chatbot asked, 'Could it be that this action involved such an emotion?', I realized for the first time that there was an emotion behind my behavior. That was impressive." In a similar vein, P8 explained: "Through conversations, I could trace back and identify the situation, the emotions it provoked, and even the underlying causes. That was very helpful." This process of exploring emotional blind spots not only helped participants reconstruct situations in more detail but also facilitated the resolution of some negative emotions. Participants noted that reflecting on the causes of their emotions through dialogue allowed them to accept and reframe those experiences. For example, P9 shared: "Talking with the emotion helped me revisit the background and the situation where I felt it. When I was angry, I usually focused only on the anger itself. But by looking at the surrounding context, the anger eventually faded." Similarly, during the sixth-week interview, P6 reflected on how his perspective had shifted over time: "At first, when the emotion asked, 'How can I help you?', I thought, 'Stress is just stress, how could it help me?' But now I feel a sense of relief that I managed to overcome the situation, and I see it as having helped me cope." These accounts suggest that dialogues with personified emotions

can foster contextual awareness, enabling participants to reinterpret the circumstances of negative emotions and, in some cases, alleviate them.

5.2.2 Uncovering Coexisting Emotions beyond the Chosen Label. In addition to fostering awareness of the contextual background of emotions, participants identified coexisting emotional elements within the same event—revealing emotions that went beyond the single label initially assigned. For instance, P6 reflected that while she conversed with a stress persona, she realized that alongside stress, she had also felt a sense of relief for managing the situation: "Sometimes it wasn't just one emotion I felt. I could recognize that I was both reassured and worried at the same time, and the dialogue helped me see that." Similarly, P8 explained: "At the time, I only thought of it as gratitude, but in that gratitude, I also felt admiration and respect—like wanting to live as diligently as that person. The conversation helped me recognize why I felt that way." These mentions illustrate how dialogue with personified emotions enabled participants not only to understand situational contexts but also to revisit coexisting, layered emotions that had been overlooked. Moreover, some participants (P4, P5, P6, P8) reported that dialogues provided them with strategies for recalling and expressing emotions in the future. These insights often emerged from agent prompts that guided users to focus on concrete aspects of their experiences. For example, during the early weeks of the study, P5—who had initially found it difficult to describe her emotions—asked: "It's so hard to describe emotions. What would help me recall them better?" The agent suggested recalling environmental cues such as the air, sunlight, or colors present at the time. P5 reflected: "When the agent guided me step by step through the environment, the memory came back naturally. This way of remembering worked well for me." She further noted that this approach not only helped in that specific dialogue but also became a recurring strategy for generating and personifying new emotions later in the study.

6 DISCUSSION AND DESIGN IMPLICATION

Prior LLM-based emotional journaling systems have primarily supported reflection through single-session writing or conversational prompts, often treating emotional entries as isolated records tied to specific moments. As a result, revisiting the same emotion over time or examining how its meaning evolves across contexts has been less central in these systems. Our findings demonstrate that Moodialogue transforms emotion recording into an ongoing reflective practice. Through the personification of emotions and engagement in both single and repeated dialogues, participants were able to uncover previously overlooked contexts and coexisting emotions. They also reframed positive and negative experiences and integrated past and present states into a more sustained understanding of their emotions. These insights highlight the value of designing systems that facilitate post-recording interactions, enabling users to revisit and reinterpret emotions over time. Such systems can surface contextual elements and meanings that were not recognized at the moment of the initial experience, thereby fostering deeper and more enduring emotional reflection.

6.1 Evolving Personified Emotions by Uncovering Emotional Blind Spots

The act of recording emotions has long been recognized as more than simple documentation, offering opportunities for self-understanding and reflection [1, 10, 62, 66]. Cognitive–behavioral research shows that expressing emotions in one's own language can help alleviate their intensity [43, 72]. Our study extends this understanding by demonstrating that personified emotions created in Moodialogue can take on new meanings through subsequent dialogues. Participants reported that they came to see emotions from broader perspectives and discovered underlying causes or coexisting emotions that had been obscured by more intense emotions. This finding highlights the value of personification in supporting repeated engagement with recorded emotions. Rather than remaining static one-time entries, emotions could be re-invoked and further explored through dialogue, allowing new dimensions of meaning to emerge over time. Unlike prior emotional journaling systems that primarily support reflection within a single recording or dialogue session, Moodialogue enabled participants to repeatedly re-engage with the same emotion over time. This repeated engagement with a personified emotion was central to the emergence of emotional blind spots and evolving interpretations observed in our findings. Importantly, these patterns emerged not from more detailed prompts or one-time conversational depth, but from Moodialogue's design that stabilizes emotions as re-engageable interactional entities over time.

However, Moodialogue currently does not allow insights gained from dialogue to be incorporated back into the persona itself. As a result, emotions remained fixed representations, limiting opportunities for them to evolve alongside users' self-understanding. This suggests the need for systems that allow emotions to progressively evolve, reflecting users' reflections and growth over time. Participants expressed interest in being able to update personas with newly discovered insights. Beyond simple profile editing, long-term engagement could be supported through mechanisms such as gamification [56]. For example, Once users engage in sustained dialogue with a particular emotion, the system could mark the emotion as having "grown," offering opportunities to refine its persona based on insights from conversations. Such mechanisms can be considered when designing mood-tracking applications that accumulate emotional records over the long term. By illustrating how emotions develop through repeated engagement, these features can help users notice shifts in how they understand or relate to their emotions. As a consequence, they can support more sustained and longitudinal forms of reflection.

Participants also described noticing secondary or coexisting emotions during their conversations—emotions that had been overshadowed by more dominant feelings. This aligns with memory research showing that people tend to recall only the most intense emotional component of an event [74], even though lived experiences encompass multiple blended emotions [33]. To translate this insight into system behavior, future designs could use LLM-based analysis to detect these coexisting emotions within the dialogue. Once the session ends, the system could be designed to offer a prompt that helps users link these newly surfaced emotions back to previously recorded ones. For example, providing a question such

as *"Would you like to name the additional emotion you noticed in this event?"* can encourage users to integrate hidden or secondary emotions into the original entry. Through this process, an emotional chat–journaling system can transform previously overlooked emotional elements into meaningful resources for ongoing reflection and future journaling. Such design directions can support the development of mood-tracking systems that capture the layered and evolving nature of emotional experience. In addition, it also enable tools aimed at facilitating emotional understanding to offer richer and more nuanced reflective opportunities.

6.2 Savoring and Cultivating Positive Feeling in Everyday Emotional Reflection

Balancing positive and negative emotions is crucial in everyday reflection. Fredrickson and Losada [16] suggest that approximately 2.9 positive experiences are needed to offset a single negative one for emotional flourishing. This underscores the importance of not only acknowledging negative emotions but also intentionally recognizing and savoring positive ones. Our findings revealed that participants derived satisfaction from savoring positive emotions through Moodialogue. In contrast, negative emotions were often reframed—such as exercise-related exhaustion being reinterpreted as pride, or anxiety as preparation for the future [18, 71].

In our field study, six participants found Moodialogue particularly well suited for recording and reflecting on negative emotions. While they valued positive emotions as sources of enjoyment, many struggled to sustain meaningful dialogue with them. Participants noted that repeated entries about everyday positive experiences often led to interactions that felt repetitive or superficial. In contrast, three participants reported benefits from recording positive emotions, and most did not express discomfort when focusing on negative ones. However, because retrospective journaling often foregrounds the most intense emotions—commonly negative [74]—there is a risk of reinforcing a skewed emphasis on negative affect. To mitigate this, mood tracking system should be designed to help users maintain a balance between positive and negative emotions. Such balance can be supported by encouraging users to identify positive meanings even within negative experiences [22] and to reflect on ways of creating future positive experiences [53]. For example, dialogues with positive emotions could adopt the metaphor of "making an appointment," in which the emotion agent suggests: "It feels like we should meet again—when and how would you like that to happen?" Considering such design elements can help users engage with a mood-tracking app not only as a tool for recording emotions but also as a means to translate positive emotions into concrete plans for future activities [19, 64].

In addition, subtle, everyday emotions may acquire meaning only after some time has passed [23]. Retrospective reflection can reveal hidden significance in these quieter emotions, although present emotional states may bias recollection [20]. To support this, when designing chat-based emotional journaling systems, it may be valuable to consider incorporating preparatory activities such as brief breathing exercises [41] or "observing thoughts" meditation [38]. These practices can help users surface overlooked aspects of past experiences and formulate questions to bring into dialogue with the personified emotion. Moreover, to deepen the overall reflective process, it may be worth considering the inclusion of a feature that allows users to record these "questions for the emotion" alongside the event title before starting the conversation. This would enable the system to prompt the agent to engage more directly with users' specific concerns. Collectively, these design opportunities suggest that Moodialogue and similar systems should not only help users savor positive emotions and reframe negative ones. They should also support the preparation, balance, and continuity of reflective dialogue. In doing so, such systems can expand emotional blind spots, foster sustained reflection, and scaffold future-oriented meaning-making.

7 LIMITATIONS AND FUTURE WORK

Regarding Moodialogue and the Agent System: In this study, participants tended to use Moodialogue not to capture the full trajectory of emotions throughout the day, but rather to focus on the most memorable emotions and events during moments of leisure, engaging in dialogue and transforming them into journal entries. This approach enabled deep reflection on specific moments, but it also presented limitations: it did not capture the diversity of emotions experienced across the day, and the personification process prior to dialogue was sometimes perceived as lengthy. Future work will explore the integration of lightweight, in-situ labeling mechanisms that allow users to briefly record emotion types and contexts during daily activities, which can then serve as anchors for more elaborate dialogue and journaling during reflective moments. Moreover, adjustments to the tone of personified emotions are needed so that they feel less like third-party interlocutors and more like companions who shared the event. To further support longitudinal reflection, we also envision incorporating retrieval-augmented generation (RAG) to reference and compare past contexts, thereby fostering a sense of continuity and sustained relationship with personified emotions. In addition, because the system relies on a commercial LLM, there is potential for interpretive bias in generated emotional tones and privacy risks related to storing personal emotional data. While we currently follow the API provider's data governance policies, we are developing a locally deployable version to ensure on-device processing and stronger user control.

Regarding Scope: Although *Moodialogue* incorporated design features inspired by cognitive–behavioral therapy techniques such as affective labeling and distanced self-talk, this study did not aim to evaluate its therapeutic efficacy. Instead, our focus was on exploring user experience and the design opportunities afforded by such interactions. Future research should examine therapeutic potential more directly, particularly when deploying the system with emotionally vulnerable populations. To mitigate possible risks, we plan to consult with clinical psychologists and psychiatrists in both system design and study protocols. Our sample size was intentionally small to support in-depth qualitative inquiry. However, all participants were in their twenties, which limits the diversity of emotional experiences captured in this study. These methodological and demographic constraint reduce the generalizability of our findings. In future work, we plan to recruit a larger and more age-diverse participant group to improve generalizability and validation. We also aim to incorporate validated instruments (e.g., PSS-10, PHQ-9)

to triangulate self-reported experiences with behavioral or clinical indicators. This approach will enable a clearer assessment of emotional distancing, reflection, and regulation outcomes.

8 CONCLUSION

This study introduced Moodialogue, a mobile journaling system that enables users to record, personify, and converse with their emotions. Through a six-week field study, we found that Moodialogue shifted emotion recording from a one-time act of documentation into a process of sustained reflection. By engaging in repeated dialogues with personified emotions, participants were able to recognize hidden causes, uncover coexisting emotions, and reinterpret both positive and negative experiences in more constructive ways. Our findings highlight three main contributions. First, personification supported emotional awareness and expression by prompting users to define, classify, and visualize their emotions, while also revealing the limitations of representing only dominant or single emotions. Second, dialogue with personified emotions uncovered blind spots, surfacing overlooked contextual elements and layered affective states, thereby broadening participants' perspectives and reframing their experiences. Third, the study offers design implications for future emotion-support systems, including the need to support evolving emotion personas, balance between positive and negative affect, and continuity across past, present, and future reflections. Taken together, these insights suggest that designing systems which integrate post-recording interactions—rather than focusing solely on immediate mood logging—can play a vital role in helping people cultivate a deeper, more enduring understanding of their emotions. By allowing emotions to evolve through iterative reflection, such systems hold the potential to scaffold emotional growth, encourage meaning-making, and ultimately support users in weaving their daily affective experiences into broader life narratives.

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